Homework 2: Redlining and environmental injustice

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In this project we will examine environmental injustices in historically redlined communities by:

- Mapping historically redlined communities
- Comparing the relationship between Low income %, Particulate percentile, and Low Lifespan percentile with HOLC grades
- Examining the relationship between bird observations and HOLC grades

I'm going to start by loading in the neccesary packages

```
pacman::p_load('tidyverse',
               'sf',
               'here',
               'tmap',
               'kableExtra',
               'patchwork',
               'ggthemes')
```

Data Aggregation

Read in the HOLC data, EJscreen, and GBIF datasets

```
holc <-  sf::st_read(here::here("data", "mapping-inequality", "mapping-inequality-los-ang

ejscreen <- sf::st_read(here::here("data",
                                   "ejscreen",
                            "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"),
                        quiet = TRUE)

# Lets filter that to be a bit more specific

california <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# Filter to LA and surrounding counties

la_county <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County",
                                 "Orange County",
                                 "San Bernardino County")) %>%
```

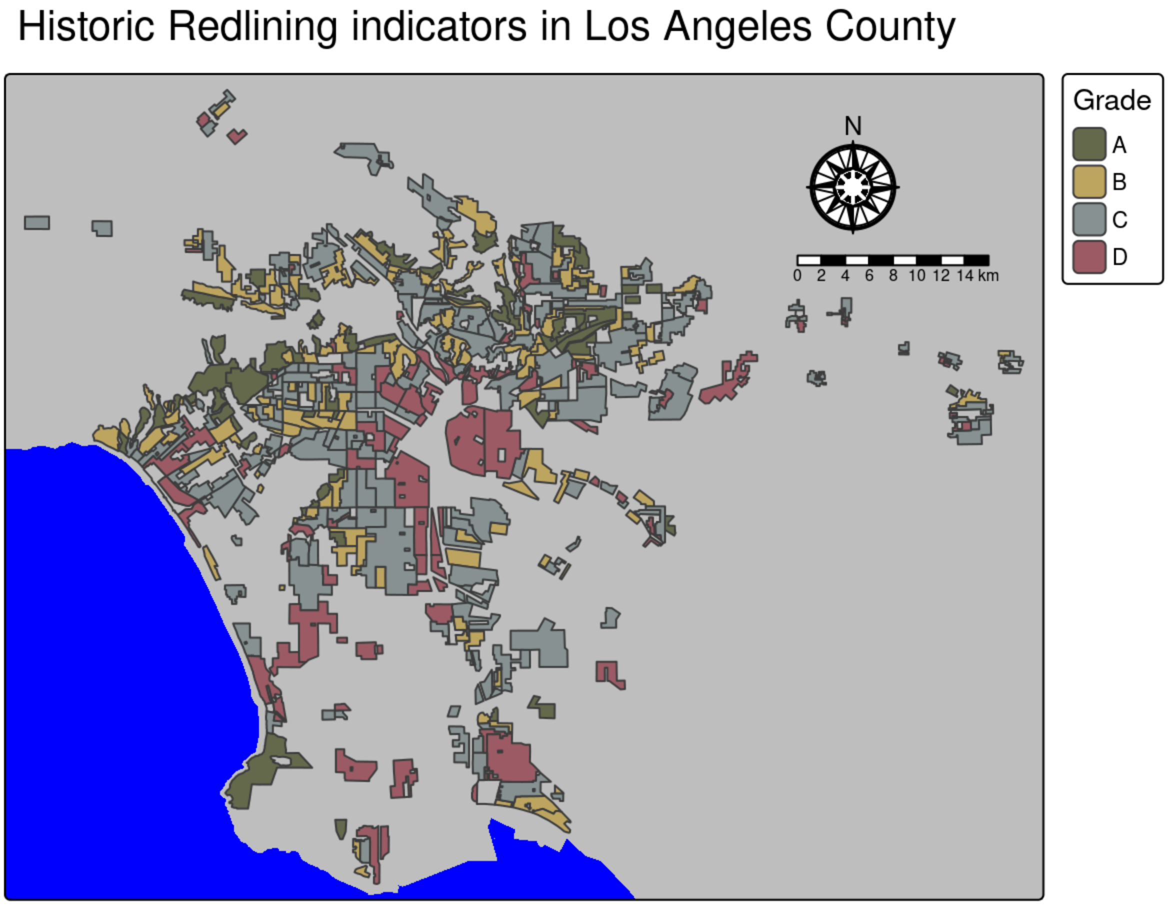
```r
  # These counties make up our area
  st_transform(crs = st_crs(holc)) %>%  # Making the CRS match the other datasets
  filter(AREALAND != 0) # Turns off the coast, making plotting prettier

gbif <-  sf::st_read(here::here("data",
                                "gbif-birds-LA",
                                "gbif-birds-LA.shp"),
                     quiet = TRUE) # Bird data
```

Part One

1.1

I am now going to make a map of the HOLC redlining.

```r
tm_shape(la_county, bbox = st_bbox(holc))+ # We set the bbox as holc
  tm_crs(4326)+
  tm_polygons(fill = 'grey',lty = 'blank')+ # Fill for my base map
  tm_title('Historic Redlining indicators in Los Angeles County')+
  tm_layout(bg.color = 'blue')+ # Fill for the Water
tm_shape(holc)+
  tm_polygons(fill = 'grade',
              fill.scale = tm_scale(values = c('#646D4E',
                                              '#C1A863',
                                              '#879595',
                                              '#9E5D67' )),
              fill.legend = tm_legend(title = 'Grade', na.show = FALSE))+
  # Took na's out of legend
tm_compass(type = 'rose', position = c('right','top'), size = 3)+ # Compass Rose
tm_scalebar(position = c('right','top'))+
  tmap_options(component.autoscale = FALSE) # Turn off autoscale
```



I set component.autoscale to false so i could be more in control of the components

1.2

I want to make a table that summarizes the percentage of census block groups that fall in each grade

I first need to join `la_county` and `holc`

```
# I have to make the HOLC Data have a valid st
holc <- st_make_valid(holc)

la_holc_ej_join <- st_join(holc, la_county) # Join my two datasets

la_holc_ej_join <- st_drop_geometry(la_holc_ej_join) # Drop the geometry

# Now I'm going to take a proportion of the census block groups that fall in to each grad

percentage_grade_df <- la_holc_ej_join %>%
  group_by(grade) %>%
  summarise(num_of = n()) %>% # Find sum of counts
  mutate(grade_percentage = round(x= (num_of / sum(num_of)) * 100,
                                  digits = 2)) %>%
  # This takes the average of the sum
  ungroup()
```

```
# And now I can make that table

kbl(percentage_grade_df,
      col.names = c("Grade", "Count", "Percentage (%)") ) %>%
    kable_classic(full_width = F, html_font = "Cambria") %>% # Turn off full width
  # Table, use Cambria to make it look like its out of a paper
  footnote(number = "Table 1. Census Block Group Percentage by HOLC Grade")
```

Grade	Count	Percentage (%)
A	449	7.03
B	1239	19.40
C	3058	47.87
D	1346	21.07
NA	296	4.63

[1] Table 1. Census Block Group Percentage by HOLC Grade

1.3

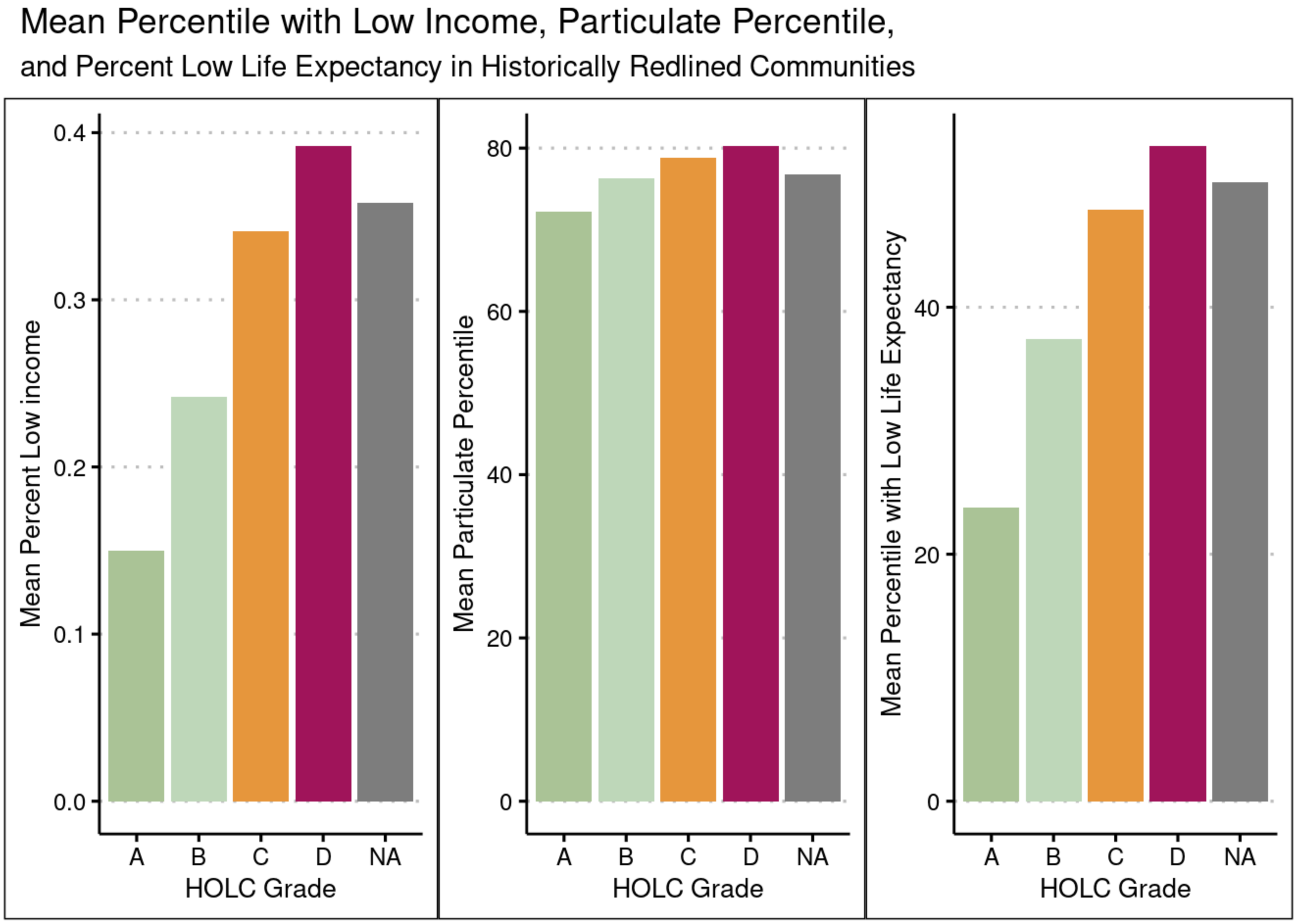
I want to make plots to see how redlining is related to some environmental justice issues

First I need to take means of my variables of interest

```
ej_mean_data <- la_holc_ej_join %>%
  group_by(grade) %>%
  summarise(mean_low_income = mean(LOWINCPCT),
            mean_particulate_percent = mean(P_PM25),
            mean_low_life_percent = mean(P_LIFEEXPPCT, na.rm = TRUE))
# Here I used the summarise function to get a bunch of means at once.
```

```
 p1 <- ggplot(ej_mean_data, aes(x = grade,
                               y = mean_low_income,
                               fill = grade))+
  geom_col()+ # This makes a bar graph with x and y axes
  scale_fill_manual(values = c('#ABC798',
                              '#C1DABD',
                              '#E6983F',
                              '#A0185A' ))+
  labs(x = 'HOLC Grade',
       y = 'Mean Percent Low income')+
  theme_clean()+
  theme(legend.position = 'none')

 p2 <- ggplot(ej_mean_data, aes(x = grade,
                               y = mean_particulate_percent,
                               fill = grade))+
  geom_col()+
  scale_fill_manual(values = c('#ABC798',
```

```
                              '#C1DABD',
                              '#E6983F',
                              '#A0185A' ))+
 labs(x = 'HOLC Grade',
      y = 'Mean Particulate Percentile')+
  theme_clean()+
 theme(legend.position = 'none')

p3 <- ggplot(ej_mean_data, aes(x = grade,
                               y = mean_low_life_percent,
                               fill = grade))+
 geom_col()+
 scale_fill_manual(values = c('#ABC798',
                              '#C1DABD',
                              '#E6983F',
                              '#A0185A' ))+
 labs(x = 'HOLC Grade',
      y = 'Mean Percentile with Low Life Expectancy')+
  theme_clean()+
 theme(legend.position = 'none')


(p1|p2|p3)+ # Use patchwork to stitch together
  plot_annotation(
 title = 'Mean Percentile with Low Income, Particulate Percentile,',
 subtitle = 'and Percent Low Life Expectancy in Historically Redlined Communities')
```



1.4

Plot summary: In these plots we can see that each variable (Percent low income, Percentile of particulates, and Percentile low life expectancy) peaks in neighborhoods given a D holc grade. This indicates that these communities being considered less desireable led them to not having investment. This in turn led to worsening environmental conditions, lower life expectancy, and concentration of low income communities which can lead to a whole host of other issues.

Part Two

2.1

I want to create a plot of percent of bird observations in each HOLC grade

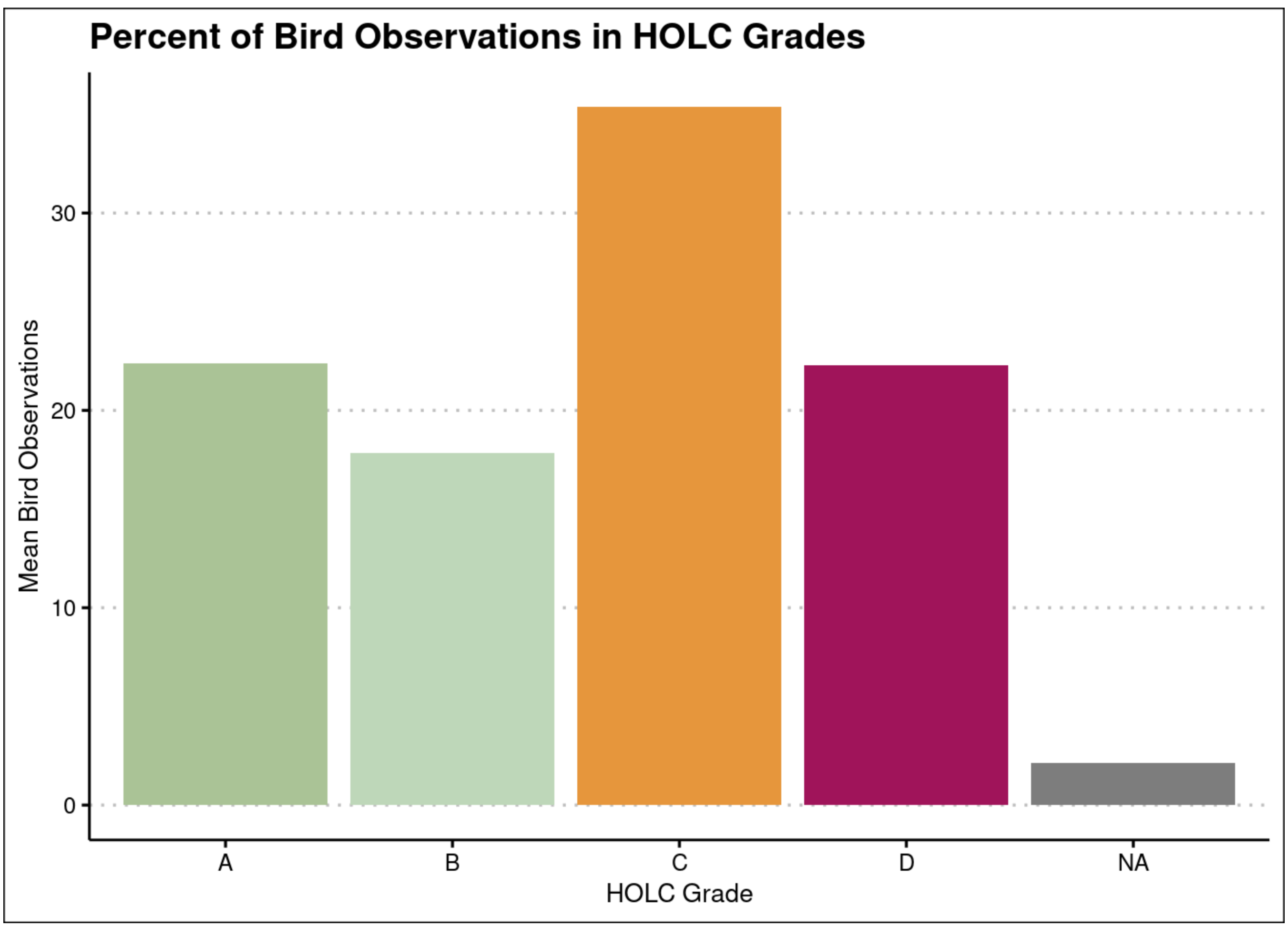
The first thing I need to do is merge `gbif` and `HOLC`

```
sf_use_s2(FALSE) # Despite the crs's matching and both being valid this next bit was givi


holc_bird_df <- st_join(holc, gbif)
```

```
holc_bird_df <- holc_bird_df %>%
  group_by(grade) %>%
  summarize(num_of = n()) %>%
  mutate(grade_percentage = round(x= (num_of / sum(num_of)) * 100,
                                  digits = 2)) %>%
  ungroup()
```

Now I can make a bar plot comparing bird counts and HOLC Grade

```
ggplot(holc_bird_df, aes(x = grade, y = grade_percentage))+
  geom_col(aes(fill = grade))+
  scale_fill_manual(values = c('#ABC798',
                               '#C1DABD',
                               '#E6983F',
                               '#A0185A' ))+
  labs(x = 'HOLC Grade',
       y = 'Mean Bird Observations',
       title = 'Percent of Bird Observations in HOLC Grades')+
  theme_clean()+
  theme(legend.position = 'none')
```



2.2

Summary:

In this plot we see, despite lack of investment, bird communities seem to thrive in C - grade areas. This indicates that bird abundance may either be geographically linked to the same areas given a C grade. Another plausible reason for this correlation is that certain bird species (generalists) are more suited to urban conditions. This plot was particularly intersting because the capstone I just worked to propose actually deals with the abundance of birds in different GAP statuses, to determine where funding would be used besr. I see an intersection in this plot where we have an area that has more birds than others, despite the lack of funding, which historically was thought to be 'undesireable'.

References

Data

United States Environmental Protection Agency. 2015. EJSCREEN. Retrieved: November 5th 2025 from epa.gov

Nelson, Robert K., LaDale Winling, et al. "Mapping Inequality: Redlining in New Deal America." Edited by Robert K. Nelson. American Panorama: An Atlas of United States History, 2023. Retrieved: October 16th 2025 from https://dsl.richmond.edu/panorama/redlining

GBIF: The Global Biodiversity Information Facility (2021) GBIF. Retrieved: October 16th 2025 from https://www.gbif.org/